An Integrated Multivariate Approach to Analyzing the Physicochemical Properties of Portuguese Vinho Verde Wines

I. INTRODUCTION

Vinho Verde are Portugal wines, aptly named after the region it originates from. It is a young wine, meaning it is meant to be drunk three to six months after harvesting. Not all wine is created equal even if it goes through the same processes, some wines are considered to be of higher quality. This wine quality may be influenced by certain physiochemical characteristics that make up wine. Furthermore, these physiochemical characteristics can be influenced by wine color. In order to see exactly how physiochemical characteristics can influence wine quality, characteristics of red and white wine, and summarize patterns that can be found in wine, multivariate statistical methods can be used.

The purpose of this study is to investigate if there are differences in the physiochemical characteristics that make up red and white wine, and if there are any underlying patterns that can be found within the characteristics.

It should also be noted that since the data used is merely observational, the results from this study will only indicate associations rather than causal relationships.

II. DATASET

This study uses the UCI Wine Quality Dataset, which has two separate CSV files for red and white wine. The combined dataset has 6,497 observations, with 1,599 red wine samples and 4,898 white wine samples. The dataset contains 11 physicochemical variables and a wine-quality score which ranges from 0 to 10 and is based on sensory evaluations. The wine color variable identifies the samples as red or white. Specifically this dataset has the following variables:

Variable	**Type**	**Description**
fixed acidity	Continuous	Fixed acidity
volatile acidity	Continuous	Volatile acidity

citric acid	Continuous	Citric acid
residual sugar	Continuous	Residual sugar
chlorides	Continuous	Chloride concentration
free sulfur dioxide	Continuous	Free sulfur dioxide
total sulfur dioxide	Continuous	Total sulfur dioxide
density	Continuous	Density
pH	Continuous	pH
sulphates	Continuous	Sulphates
alcohol	Continuous	Alcohol concentration
quality	Integer	Sensory quality score
color	Categorical	Red or White

III. RESEARCH QUESTIONS AND HYPOTHESES

Research Question 1 – MANOVA:
Do red and white wines differ significantly in their overall physicochemical characteristics when several characteristics are considered simultaneously?

The characteristics chosen for this MANOVA were: total sulfur dioxide, alcohol, residual sugar, and pH.

H_0: The mean vector of the selected physicochemical variables is the same for red and white wines
H_1: The mean vectors differ between red and white wines

Research Question 2 – PCA:
Can the 11 physicochemical measurements of wine be summarized using a smaller number of principal components while retaining a substantial proportion of the total variability?

Research Question 3 – EFA:
What underlying latent dimensions or factors explain the relationships among the physicochemical characteristics of wine?

IV. METHODOLOGY

4.1 Data Preparation

In order to prepare the data, the individual red and white wine quality sets were loaded in, and a color categorical variable was created to identify the samples as either red or white wine. Afterwards, the two datasets were concatenated and dataset structure and descriptive statistics were examined. The combined dataset was checked to see if there were missing values and duplicate rows. While there were no missing rows, there were 1,177 exact duplicate rows which were retained.

4.2 Exploratory Data Analysis

Potential univariate outliers were determined using the interquartile range (IQR) method, and potential multivariate outliers were determined through Mahalanobis distance. Graphs were created for further exploratory data analysis. A histogram was created to show distributions of major physiochemical variables, and a correlation matrix was used to identify relationships between quantitative variables. Additionally, histograms and boxplots were used to compare physiochemical variable distribution of red and white wines.

4.3 Multivariate Analysis of Variance (MANOVA)

MANOVA was conducted using wine color as both the independent variable and groups. The following phsyiochemical characteristics were selected to be the dependent variables: total sulfur dioxide, alcohol, residual sugar, and pH. Levene's tests were used in order to determine the homogeneity of variance for the dependent variables, and correlation among the dependent variables were also examined. Afterwards MANOVA was conducted with Pillai's trace reported as the multivariate test statistic. Follow-up univariate analyses were also done to examine each dependent variable individually, ANOVA was conducted with eta-squared reported as an effect-size measure.

4.4 Principal Component Analysis (PCA)

First, as the variables all had different scales, they were all standardized before performing PCA. PCA was then fitted using the standardized data. Eigenvalues, proportions of explained variance, cumulative explained variance, a scree plot, and component loadings were all used to determine and interpret the retained components.

4.5 Exploratory Factor Analysis (EFA)

Factorability was assessed using the Kaiser--Meyer--Olkin (KMO) measure and Bartlett's test of sphericity were used to assess factorability. To determine which factors would be retained, eigenvalues, a scree plot, and parallel analysis were used. A three-factor EFA was fitted using the minimum residual (minres) extraction method and Promax oblique rotation. Furthermore, factor loadings, communalities, and uniqueness values were inspected to interpret the factors and assess the EFA.

4.6 Statistical Software

Python was used in a Google Colab notebook. The notebook uses pandas and NumPy for data handling, Matplotlib and Seaborn for visualization, SciPy and statsmodels for statistical testing, scikit-learn for standardization and PCA, and factor_analyzer for KMO, Bartlett's test, and EFA.

V. Results

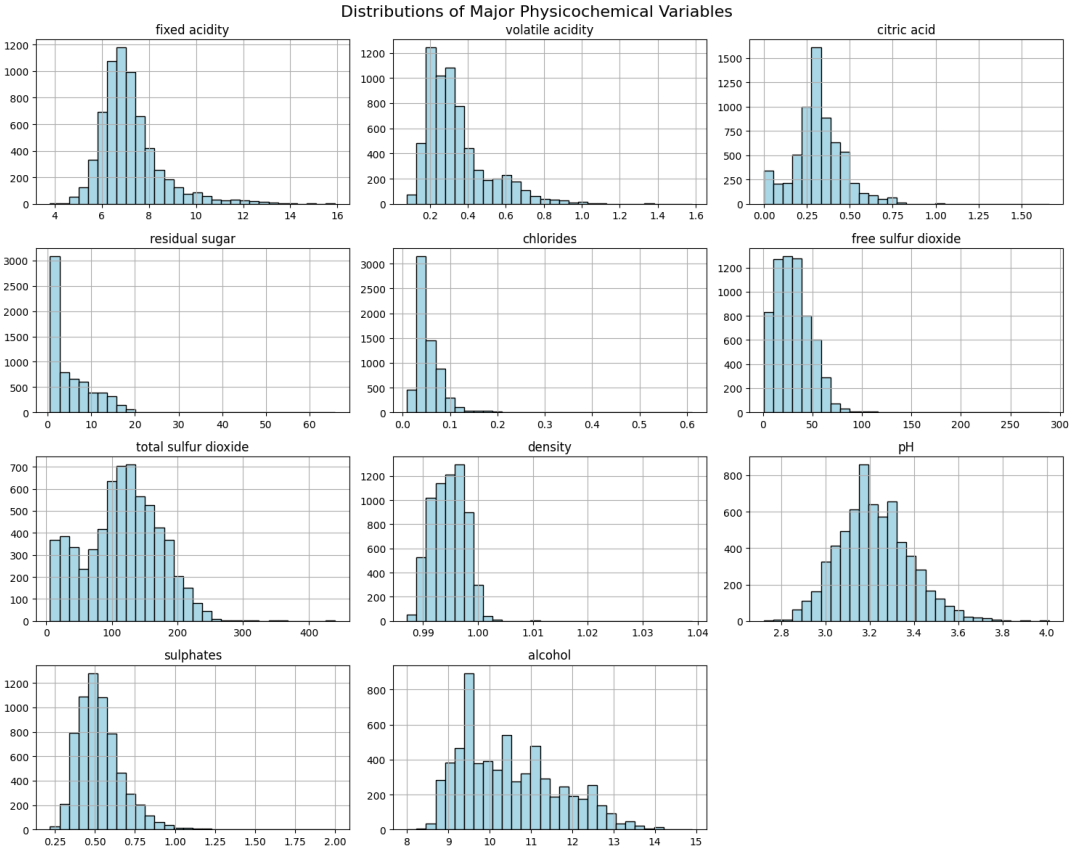
5.1. Exploratory Data Analysis

The combined dataset contained 6,497 observations with 1,599 red wine samples and 4,898 white wine samples. s and 13 columns. It had no missing values and 1,177 exact duplicate rows, which were chosen to be retained rather than removed, as it was possible for wine to have the same characteristics.

Variable	Count	Mean	Std	Min	25%	Median (50%)	75%	Max
Fixed acidity	6497	7.215	1.296	3.800	6.400	7.000	7.700	15.900
Volatile acidity	6497	0.340	0.165	0.080	0.230	0.290	0.400	1.580
Citric acid	6497	0.319	0.145	0.000	0.250	0.310	0.390	1.660

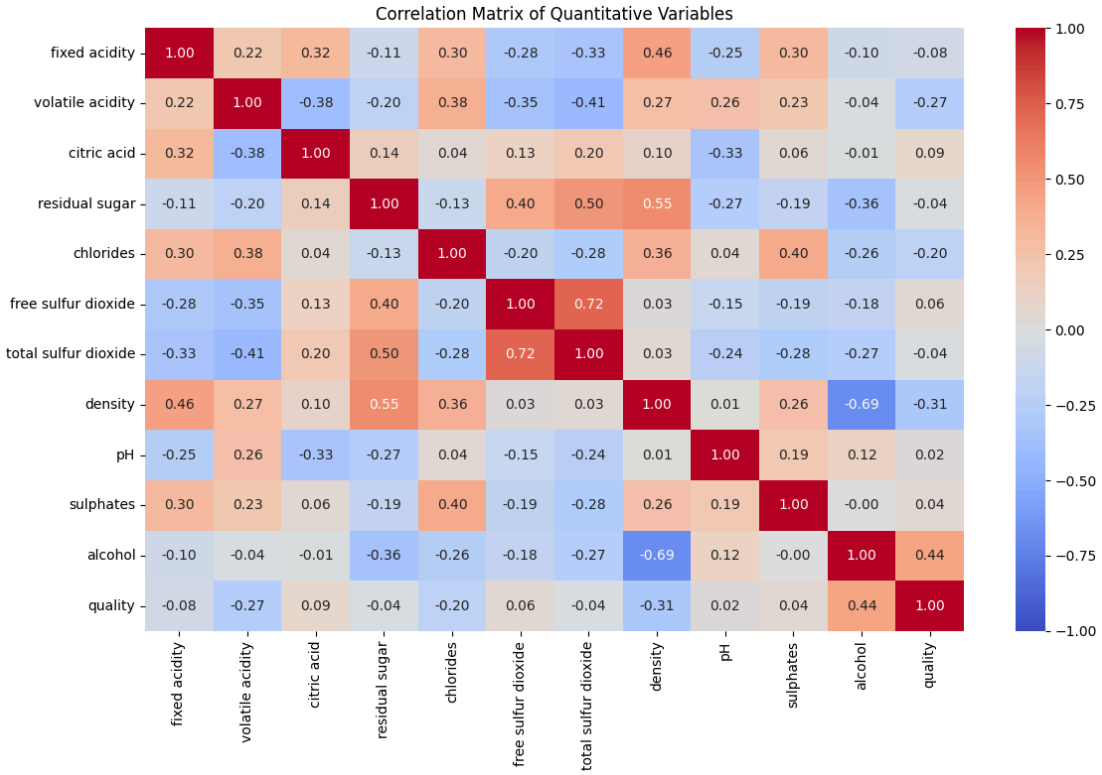
Residual sugar	6497	5.443	4.758	0.600	1.800	3.000	8.100	65.800
Chlorides	6497	0.056	0.035	0.009	0.038	0.047	0.065	0.611
Free sulfur dioxide	6497	30.525	17.749	1.000	17.000	29.000	41.000	289.000
Total sulfur dioxide	6497	115.745	56.522	6.000	77.000	118.000	156.000	440.000
Density	6497	0.995	0.003	0.987	0.992	0.995	0.997	1.039
pH	6497	3.219	0.161	2.720	3.110	3.210	3.320	4.010
Sulphates	6497	0.531	0.149	0.220	0.430	0.510	0.600	2.000
Alcohol	6497	10.492	1.193	8.000	9.500	10.300	11.300	14.900
Quality	6497	5.818	0.873	3.000	5.000	6.000	6.000	9.000

Table 1. Descriptive Statistics



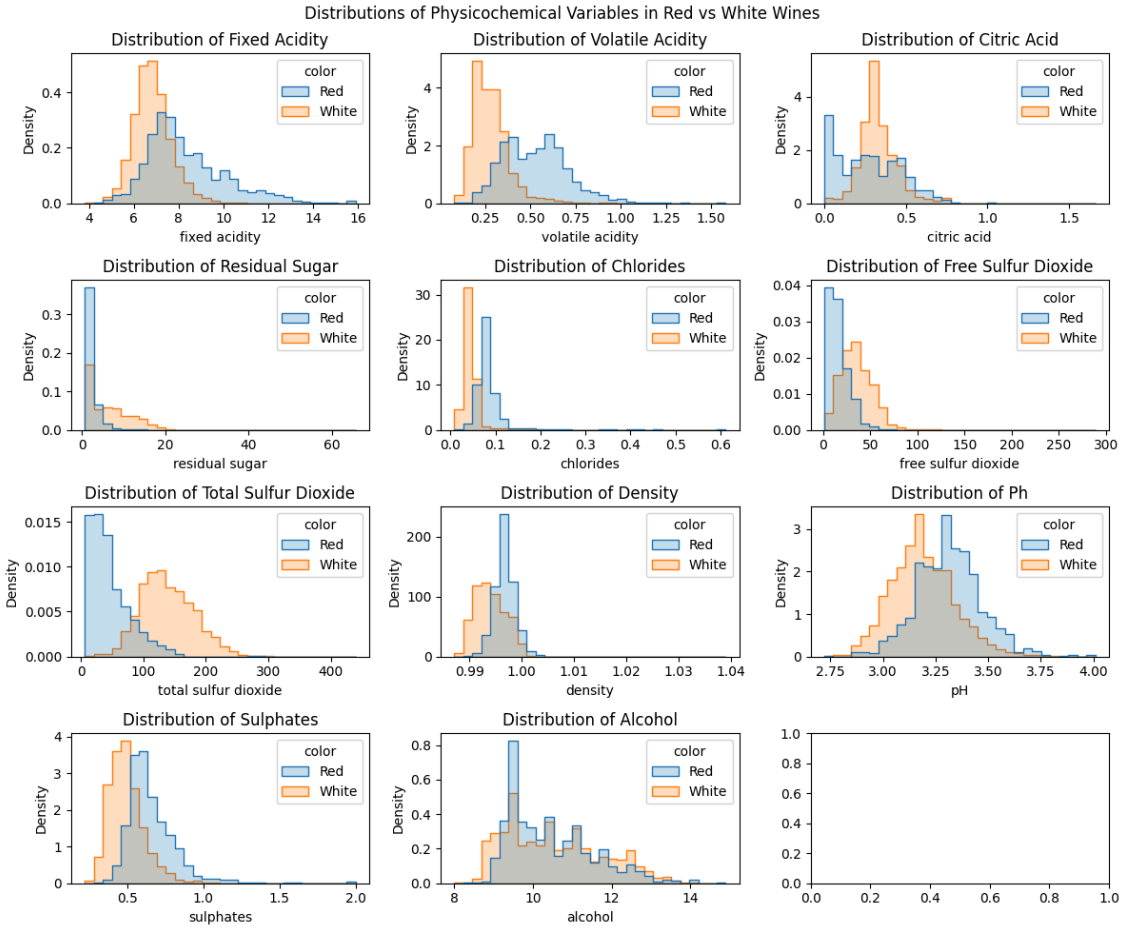
Graph 1. Histograms showing Distribution of Major Physiochemical Variables

Table 1 and Graph 1 show that both sulfur dioxide variables show substantial variability. Free sulfur dioxide has a standard deviation of 17.749 and ranges from 1 to 289, while total sulfur dioxide has a standard deviation of 56.522 and ranges from 6 to 440.

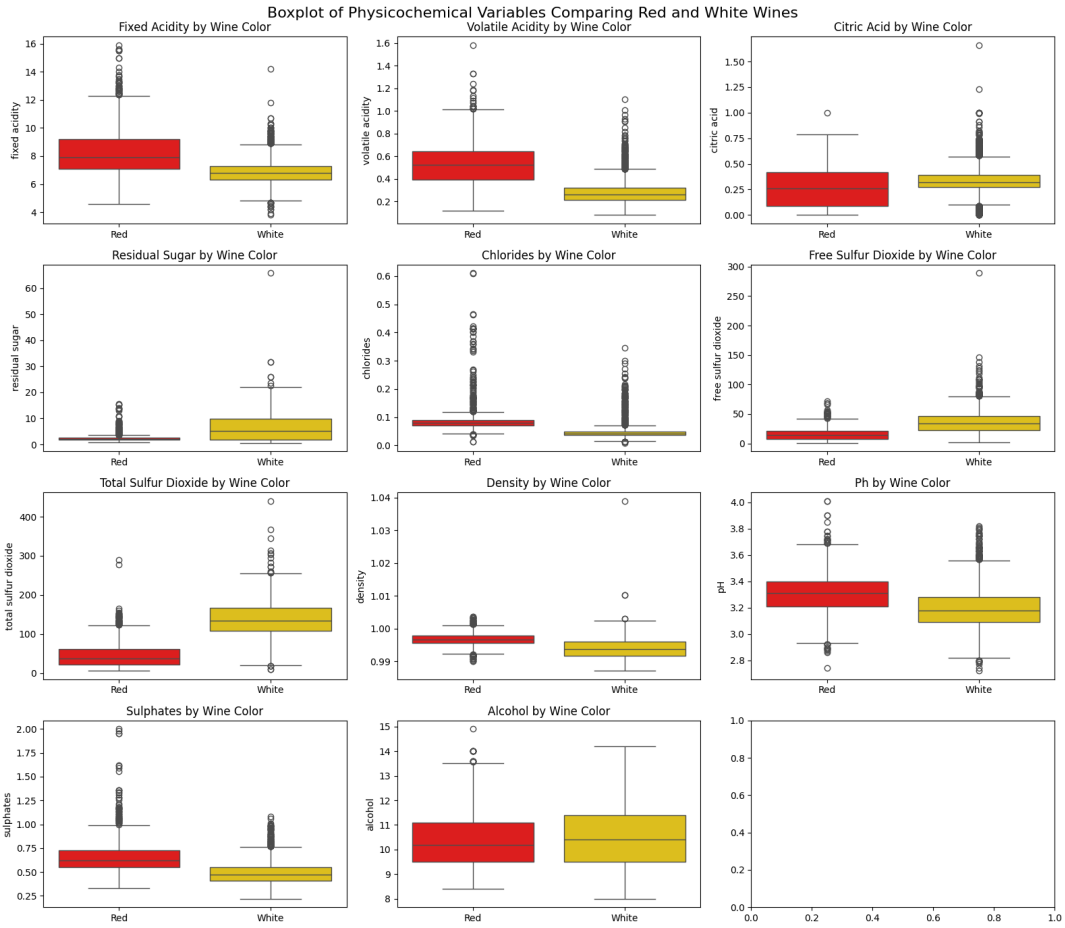


Graph 2. Correlation Matrix of Quantitative Variables

Graph 2 shows that the variables which appear to be strongly correlated are: total sulfur dioxide and free sulfur dioxide (r = 0.72), density and alcohol (r = -0.69), and residual sugar and density (r = 0.55).



Graph 3. Histogram Comparing Red and White Wines



Graph 4. Boxplot Comparing Red and White Wines

Graph 3 and Graph 4 confirms that there are visible differences in the physiochemical characteristics of red and white wines. Red wines were shown to typically have higher volatile acidity, chlorides, sulphates, and fixed acidity. Meanwhile, white wines were shown to have higher total sulfur dioxide, free sulfur dioxide, and slightly higher residual sugar.

Variable	**Potential Outliers**	**Percentage (%)**
fixed acidity	357	5.49
volatile acidity	377	5.80
citric acid	509	7.83

residual sugar	118	1.82
chlorides	286	4.40
free sulfur dioxide	62	0.95
total sulfur dioxide	10	0.15
density	3	0.05
pH	73	1.12
sulphates	191	2.94
alcohol	3	0.05
quality	228	3.51

Table 2. Univariate Outliers

The results for IQR can be found in Table 2 which contains Univariate outliers and it was also found, using Mahalanobis distance, that with a chi-square cut off of 24.725, there are 295 potential multivariate outliers, which is 4.54% of the dataset.

5.2. Multivariate Analysis of Variance (MANOVA)

The MANOVA conducted seeks to answer the research questions: Do red and white wines differ significantly in their overall physicochemical characteristics when several characteristics are considered simultaneously?

It has the following hypotheses:

H_0: The mean vector of the selected physicochemical variables (total sulfur dioxide, alcohol, residual sugar, and pH) is the same for red and white wines

H_1: The mean vectors differ between red and white wines

Variable	W-stat	p-value
total_sulfur_dioxide	181.381	8.5065e-41
alcohol	72.785	1.7802e-17
residual_sugar	1867.544	0.0000e+00
ph	0.482	4.8770e-01

Table 3. Homogeneity of Variances (Levene's Test)

Variable	Total Sulfur Dioxide	Alcohol	Residual Sugar	pH
Total Sulfur Dioxide	**1.000**	**-0.266**	**0.495**	**-0.238**
Alcohol	**-0.266**	**1.000**	**-0.359**	**0.121**
Residual Sugar	**0.495**	**-0.359**	**1.000**	**-0.267**
pH	**-0.238**	**0.121**	**-0.267**	**1.000**

Table 4. Relationships Among Dependent Variables for MANOVA (Pearson Correlation)

In Table 3, Levene's test showed that the equal-variance assumption is violated for total sulfur dioxide, alcohol, and residual sugar, but not for pH. The correlations among the four dependent variables are found in Table 4 are: total sulfur dioxide and residual sugar (r=0.495), alcohol and residual sugar (r=-0.359), total sulfur dioxide and alcohol (r=-0.266), residual sugar and pH(r=-0.267), total sulfur dioxide and pH (r=-0.238), and alcohol and pH (r=0.121).

Statistic	Value	Num DF	Den DF	F Value	Pr > F
Wilks' lambda	0.423616	4.0	6492.0	2208.301071	0.0
Pillai's trace	0.576384	4.0	6492.0	2208.301071	0.0
Hotelling-Lawley trace	1.360629	4.0	6492.0	2208.301071	0.0
Roy's greatest root	1.360629	4.0	6492.0	2208.301071	0.0

Table 5. MANOVA

Metric	Value
Test Statistic (Pillai's Trace)	0.5764
F-value	2208.3011
Degrees of Freedom (Num, Den)	(4.0, 6492.0)
p-value	0.0000e+00
Multivariate Effect Size (η^2)	0.5764

Table 6. Pillai's Trace

Table 5 and 6 showed that the MANOVA is statistically significant as Pillai's Trace (= 0.5764), ($F(4, 6492) = 2208.3011$),($p < 0.001$). This result indicates that wine color is significantly associated with differences in the combined profiles of total sulfur dioxide, alcohol, residual sugar, and pH. Since the MANOVA is significant, follow-up analyses can be conducted.

Dependent Variable	F-value	p-value	Effect Size (Eta-squared)	Statistical Significance
Total Sulfur Dioxide	6252.7956	< 0.001	0.4905	Significant
Alcohol	7.0677	0.0079	0.0011	Significant
Residual Sugar	899.7663	2.9594×10^{-185}	0.1217	Significant
pH	789.0495	5.9233×10^{-164}	0.1083	Significant

Table 7. Univariate ANOVAs

The ANOVA analysis in Table 7 revealed that total sulfur dioxide has the largest effect size of 0.4905. This means that wine color accounts for a substantial amount of the variance in total sulfur dioxide. Additionally, residual sugar and pH also show meaningful group differences, with effect sizes of 0.1217 and 0.1083 respectively.

In the ANOVA, alcohol is statistically significant, but its effect size is extremely small, only 0.0011. This is to be expected as due to the large number of observations, a very small difference can still lead to statistical significance. Therefore, alcohol should not be considered as an important practical contributor to the difference in red and white wine.

Overall, the MANOVA indicates that red and white wines significantly different when the chosen physiochemical variables (total sulfur dioxide, alcohol, residual sugar, pH) are considered simultaneously. Therefore, we reject the null hypothesis and accept the alternative hypothesis.

5.3. **Principal Component Analysis (PCA)**

The PCA conducted seeks to answer the following research question: Can the 11 physicochemical measurements of wine be summarized using a smaller number of principal components while retaining a substantial proportion of the total variability

Component	Eigenvalue	Proportion of Variance	Cumulative Variance
PC1	3.0303	0.2754	0.2754
PC2	2.4942	0.2267	0.5022
PC3	1.5566	0.1415	0.6436
PC4	0.9707	0.0882	0.7319
PC5	0.7200	0.0654	0.7973
PC6	0.6074	0.0552	0.8525
PC7	0.5232	0.0476	0.9001
PC8	0.5016	0.0456	0.9457
PC9	0.3371	0.0306	0.9763
PC10	0.2277	0.0207	0.9970
PC11	0.0328	0.0030	1.0000

Table 8. PCA Components with Eigenvalues, Proportion of Variance, andCumulative Variance

While there were 11 components produced, in Table 8, it can be seen that only the first three components had eigenvalues greater than one. Additionally, the first three components have a cumulative percentage of 64.36%. This means that these three components explain almost two-thirds of the standardized variance. Which is why it was decided to only retain the first three components.

Variable	PC1	PC2	PC3
Fixed acidity	-0.239	0.336	0.434
Volatile acidity	-0.381	0.118	-0.307
Citric acid	0.152	0.183	0.591
Residual sugar	0.346	0.330	-0.165
Chlorides	-0.290	0.315	-0.017
Free sulfur dioxide	0.431	0.072	-0.134
Total sulfur dioxide	0.487	0.087	-0.107
Density	-0.045	0.584	-0.176
pH	-0.219	-0.156	-0.455
Sulphates	-0.294	0.192	0.070
Alcohol	-0.106	-0.465	0.261

Table 9. Retained Components Loading

It was found that PC1 explains 27.54% of the total variance, with its strongest positive loadings shown in Table 9, this includes the following variables: total sulfur dioxide (0.487), free sulfur dioxide (0.431), and residual sugar (0.346). It also has strong negative loadings for the following variables: volatile acidity (-0.381),chlorides (-0.290), and sulphates (-0.294). Therefore, PC1 represents a sulfur dioxide and broader chemical-composition dimension, and should be named sulfur dioxide and chemical composition.

Meanwhile, PC2 explains 22.67% of the variance, with its strongest positive loadings including the following variables: density (0.584), and residual sugar (0.33). It also has a moderately strong negative loading in alcohol (-0.465). Thus, PC2 represents a density–sugar–alcohol dimension, and should be named density, residual sugar, and alcohol.

Lastly, PC3 explains 14.15% of the variance, with its strongest positive loadings including the following variables: citric acid (0.591) and fixed acidity (0.434). It also has strong negative loadings for the following variables: pH (-0.455), and volatile acidity (-0.307). Therefore, PC3 represents differences associated with citric acid, fixed acidity, pH, and volatile acidity, and should be named acid composition.

Thus, the answer to the research question is that the PCA reduces the 11 physicochemical measurements into three components, while still being able to retain 64.36% of the total standardized variance.

5.4. Exploratory Factor Analysis (EFA)

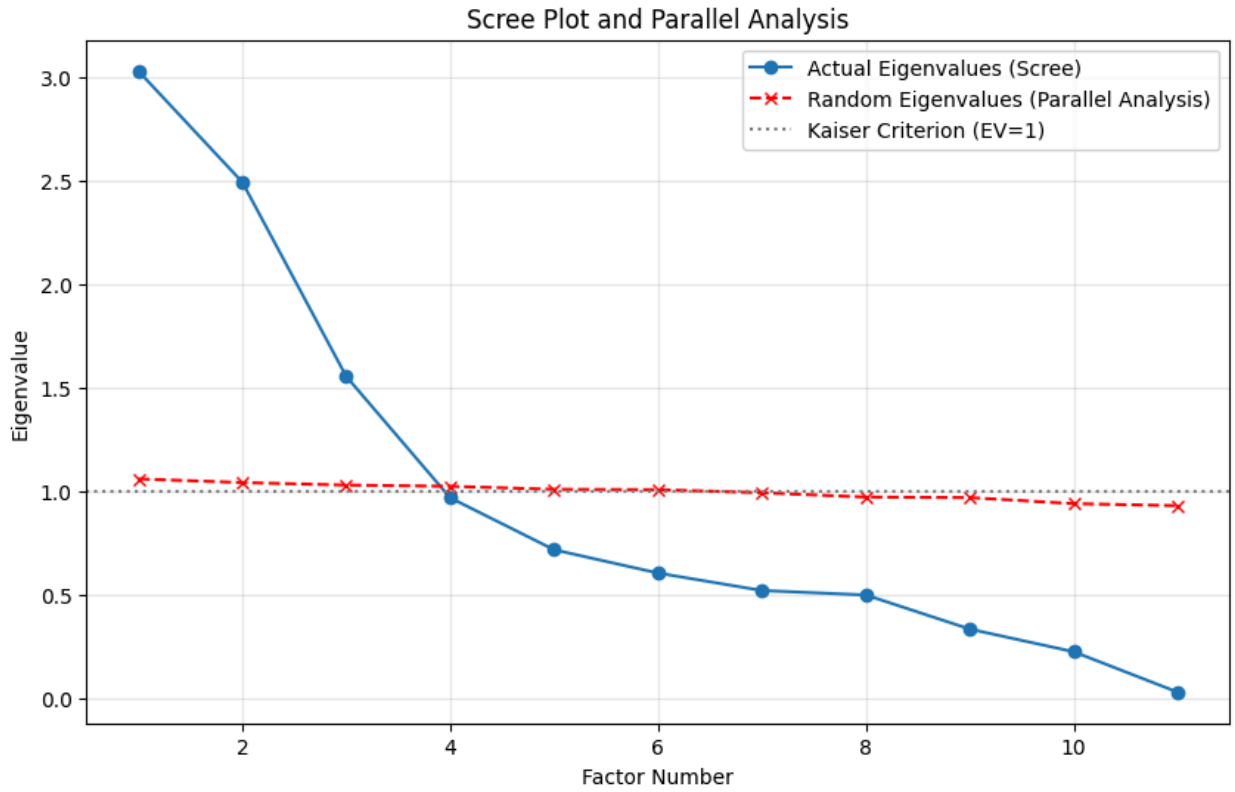
The EFA conducted seeks to answer the following research question: What underlying latent dimensions or factors explain the relationships among the physicochemical characteristics of wine?

First, Bartlett's test is highly significant, with chi squared =37094.62,and $p<.001$, which means the results support factorability of dataset, However KMO 0.405, which is relatively and low and is an indication of weak overall sampling adequacy for factor analysis. This means that the results from factor analysis should be interpreted cautiously.

Factor	Actual Eigenvalue	Random Eigenvalue
Factor 1	3.030	1.062
Factor 2	2.494	1.044
Factor 3	1.556	1.032
Factor 4	0.971	1.026

Factor 5	0.720	1.011
Factor 6	0.607	1.010
Factor 7	0.523	0.995
Factor 8	0.502	0.974
Factor 9	0.337	0.972
Factor 10	0.228	0.942
Factor 11	0.033	0.932

Table 10. Actual vs Random Eigenvalues of Factors



Graph 5. Scree Plot and Parallel Analysis for EFA

Table 10 and Graph 5 both show that of the Eigenvalues (Kaiser Criterion), only the first three factors have actual eigenvalues greater than 1.0 (3.030, 2.494, and 1.556) and that comparing the data's actual eigenvalues to those generated from random simulated data shows that the actual eigenvalues drop below the random eigenvalues at the fourth factor. Based on all of these, it was determined that using 3 factors is the most appropriate decision.

Variable	Factor 1	Factor 2	Factor 3
Fixed acidity	0.745	0.068	0.560
Volatile acidity	0.490	0.165	-0.415
Citric acid	0.091	-0.053	0.738
Residual sugar	-0.338	0.619	0.064
Chlorides	0.527	0.237	-0.018
Free sulfur dioxide	-0.599	0.297	0.056
Total sulfur dioxide	-0.724	0.360	0.106
Density	0.423	0.952	-0.017
pH	0.067	-0.025	-0.514
Sulphates	0.472	0.067	0.022
Alcohol	-0.040	-0.705	0.067

Table 11. Factor Loadings

Variable	Communalities	Uniqueness
Fixed acidity	0.874	0.126
Volatile acidity	0.440	0.560
Citric acid	0.556	0.444
Residual sugar	0.502	0.498
Chlorides	0.335	0.665
Free sulfur dioxide	0.450	0.550
Total sulfur dioxide	0.665	0.335
Density	1.085	-0.085
pH	0.270	0.730
Sulphates	0.228	0.772
Alcohol	0.503	0.497

Table 12. Communalities and Uniqueness

Based on the results from Table 11 and Table 12, the highest loadings for Factor 1 is: fixed acidity (0.745), total sulfur dioxide (-0.724), free sulfur dioxide (-0.599), chlorides (0.527), volatile acidity (0.490), and sulphates (0.472). It can be seen that this factor represents the pattern involving acidity-related variables and sulfur dioxide variables. Therefore, it should be called acidity–sulfur dioxide dimension.

Meanwhile for Factor 2, the highest loadings are: density (0.952), alcohol (-0.705), and residual sugar (0.619). Thus this factor should be called the density–sugar–alcohol dimension.

Lastly, the highest loadings of Factor 3 are: citric acid (0.738), fixed acidity (0.560), pH (-0.514), and volatile acidity (-0.415); this factor should be called the acid composition dimension. This factor represents a common pattern among citric acid, fixed acidity, pH, and volatile acidity. Therefore it should be called acid composition dimension.

To answer the research question, there are three underlying dimensions of physiochemical characteristics revealed by the EFA which are:

- The acidity acidity–sulfur dioxide dimension, which has the variables: fixed acidity, total sulfur dioxide, free sulfur dioxide, chlorides, volatile acidity, and sulphates.

- The density–sugar–alcohol dimension, which has the variables: density, alcohol, and residual sugar.

- The acid composition dimension, which has the variables: citric acid, fixed acidity, pH, and volatile acidity.

VI. Integrated Discussion

The MANOVA showed that red and white wines have significantly different combined physicochemical profiles. The specific variables used for this MANOVA analysis are: total sulfur dioxide, alcohol, residual sugar, and pH. The follow-up ANOVA analyses showed that total sulfur dioxide had the largest effect size, while residual sugar and pH also had meaningful effects. Alcohol was statistically significant but only had a very small effect size. Therefore, it can be said that the difference between red and white wines is associated with differences in specific aspects of their composition rather than a large difference in every selected variable.

The PCA showed that three components explain 64.36% of the total standardized variance. PC1, named sulfur dioxide and chemical composition, includes total and free sulfur dioxide, residual sugar, volatile acidity, chlorides, and sulphates. PC2, named density, residual sugar, and alcohol, involves density, residual sugar, and alcohol, while PC3, named acid composition, consists of citric acid, fixed acidity, pH, and volatile acidity.

The EFA also supported a three-factor structure, similar to the three components in PCA. Although it should be noted that it has a low KMO value meaning results should be interpreted cautiously. Factor 1, named acidity–sulfur dioxide dimension, includes fixed acidity, total and free sulfur dioxide, chlorides, volatile acidity, and sulphates. Factor 2, named

density–sugar–alcohol dimension, involves density, alcohol, and residual sugar, while Factor 3, named acid composition dimension, consists of citric acid, fixed acidity, pH, and volatile acidity.

There are strong commonalities that can be seen between PCA and EFA, the most prominent being the density–sugar–alcohol relationship. PCA's PC2 has a density loading of 0.584 and an alcohol loading of -0.465, while EFA Factor 2 has a very strong density loading of 0.952, a residual sugar loading of 0.619, and an alcohol loading of -0.705. Additionally, the EDA density has a moderately strong positive correlation with residual sugar, and moderately strong negative correlation with alcohol. This consistency indicates that the relationship among density, residual sugar, and alcohol is one of the underlying patterns in the dataset. While in ANOVA, alcohol only has a small effect size, residual sugar did show meaningful group differences.

Another recurring feature across the analyses is sulfur dioxide, as it contributes strongly to PC1 in Part C, and has substantial loadings on Factor 1 in Part D. Furthermore, in MANOVA, amongst the selected variables, total sulfur dioxide has the largest effect size. This indicates that sulfur dioxide measurements represent a significant source of shared variation among the wines.

Acidity is also identified in PCA and EFA. Citric acid, fixed acidity, pH, and volatile acidity contribute strongly to both PC3 and Factor 3. Therefore, the acidity-related structure is not limited to a single variable but reflects relationships among several chemical measurements.

Therefore, all together the analyses tell a consistent overall story. It can be seen that there are differences in the physiochemical profile of the red and white wines. These variations within the wines can be summarized by several recurring chemical dimensions. The most notable of these include sulfur dioxide, density–residual sugar–alcohol, and acidity.

VII. Conclusion

This study analyzed 6,497 Portuguese Vinho Verde wine observations (1,599 red wine samples and 4,898 white wine samples) using EDA, MANOVA, PCA, and EFA. EDA visualized important correlations among sulfur dioxide variables, density and residual sugar, density and alcohol, as well as distributional differences between red and white wines. Conducting MANOVA then confirmed a statistically significant difference in the combined profiles of total sulfur dioxide, alcohol, residual sugar, and pH, with total sulfur dioxide showing the largest follow-up effect size and alcohol showing only a small effect size. PCA reduced the 11 standardized measurements to three components which manage to explain 64.36% of total variance. These three components were: sulfur dioxide and chemical composition; density, residual sugar, and alcohol; and acid composition. EFA also suggested three broadly similar dimensions, specifically: acidity–sulfur dioxide dimension; density–sugar–alcohol dimension; and acid composition dimension. Although the Bartlett's Test supports this factorability, the EFA results should still be interpreted cautiously due to the resulting low KMO. All together, the

analyses and results prove that wine color is associated with differences in physicochemical profile and that several recurring patterns can summarize relationships among the measurements.

VIII. References

UCI Machine Learning Repository. (n.d.). *Wine Quality Dataset*. https://archive.ics.uci.edu/dataset/186/wine+quality